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Energy Forward Curves

4.1 Introduction

In this chapter we discuss approaches that have been put forward by a number of authors for constructing energy forward curves. Further useful discussions are provided in Gabillon (1995), Humphreys and Shimko (1997), and Leong (1997). Forwards and futures markets are often used by risk managers to hedge their risk, with liquid forward prices helping the price discovery mechanism to determine the fair value for future delivery. Prices from these markets are also the key inputs to many derivative pricing models – we will see in chapter 8 that energy derivative prices can be evaluated given the forward curve and the associated volatilities of the forward prices. As such, forward prices and volatilities play an essential role in energy markets and are as important as the level of spot prices.

The forward curve contains information about the prices an investor can lock into today for different times in the future. Forward prices allow users to lock into buying (long forward) or selling (short forward) spot energy at a certain time in the future at the forward price. Figure 4.1 illustrates three market forward energy curves. Each curve is a plot of the first 24 monthly contracts with prices taken from NYMEX.

The curve for NYMEX crude oil futures at the end of March 2000 is clearly in *backwardation* (futures prices lower than the spot price) with short dated futures trading at just under \$27 dollars, declining to \$21 for the two year contract. A price of \$21 implies that the holder of the long position can buy oil at \$21 at the maturity date of the contract, irrespective of the spot price for oil on that date. The crude oil futures curve just over a year earlier however, at the beginning of March 1999, is in *contango* (futures prices higher than the spot price) with short dated futures trading at prices below their longer dated counterparts.

The third futures curve shown in figure 4.1 is the NYMEX Henry Hub natural gas futures contract observed on the 31 March 2000 and clearly shows the strong seasonality of energy prices associated with the North American seasons. Later in this chapter we explain how forward curves can take these three shapes dependent on the level of transaction costs, the concept of a “convenience yield”, and seasonal changes in supply and demand.

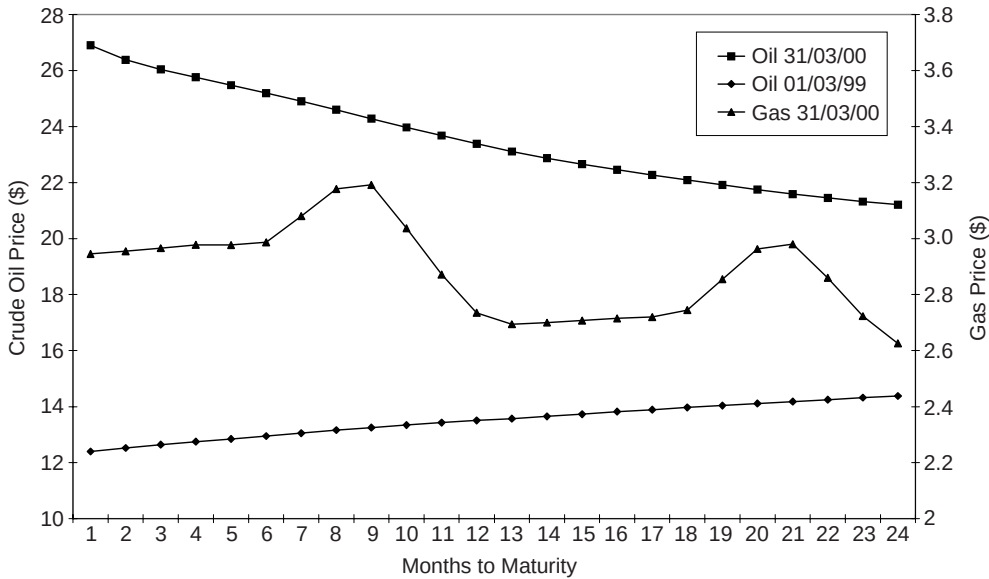


FIGURE 4.1 Market Forward Curves

Forward Curves in the Debt Markets

Forward curves are well-known and understood in the debt markets. Forward rate agreements and exchange traded futures contracts (for example Eurodollar futures contracts) are heavily traded and allow users to lock-in borrowing and lending rates for future time periods. These forward rates are easily implied from the prices of bonds and spot interest rates. For example, if the price of a 6 month pure discount bond is trading at 0.9608 and the price of a 9 month pure discount bond is trading at 0.9347 then the implied 3 month forward rate starting in 6 months time, with continuous compounding is 11.02%, given by;

$$-\frac{1}{0.25} \ln \left[\frac{0.9347}{0.9608} \right] \quad (4.1)$$

The pricing of the forward interest rate instrument is therefore determined from the current term structure of interest rates (described above by the prices of pure discount bonds). Equation (4.1) shows that forward rates are not arbitrarily set but depend on the relationship between traded instruments. The calculation also shows that the forward rate is calculated based on the observed current spot values and is not dependent on the *forecasted* level of future spot values. Leong (1997) makes the point that, for many players in the immature electricity markets, “there is not even a consensus among market

players as to what is meant by the term ‘forward curve’¹. Many participants, and even some data vendors, mistake schedules of price forecasts for forward prices. Price forecasts are predictions on the likely spot price for periods in the future and can differ widely between market participants. Forward prices, on the other hand lock in tradeable prices today for future spot transactions and as such capture the market reality. As far as possible these prices should be observed directly from the market.

4.2 Constructing Forward Curves

In the financial markets, no arbitrage models relating spot and forward prices offer a direct and reliable valuation of forward prices. This relationship is governed by arbitrage between the spot and forward contracts which is facilitated by the ability to easily buy and sell the spot instrument. However, for many energies the relationship cannot be described so simply and this will have an impact on the forward prices observed in these markets.

4.2.1 Cost of Carry Relationship

For many markets the price of a futures contract can be derived from the price of the spot instrument and the financing cost. This direct relationship results from the simple fact that the payoff to a forward sale of an asset, for example a stock, can be replicated by borrowing the purchase price in the money market, purchasing the stock, holding (or “carrying”) the stock until maturity, and then delivering the asset into the long position, using the funds received to pay off the loan. This type of arbitrage is often referred to as “cost of carry arbitrage” and leads to the following pricing formula for the s -maturity futures price at time t ;

$$F(t, s) = S(t)e^{r(s-t)} \quad (4.2)$$

where r is the financing cost, assuming continuous compounding, over the life of the forward position, $s - t$, and $S(t)$ is the asset price at time t . It is easy to see that if the market prices for the stock and the future on the stock do not satisfy equation (4.2) then there are arbitrage profits to be made. For example, if the market forward price is greater than the carry relationship ($F(t, s) > S(t)e^{r(s-t)}$), investors can borrow an amount equal to the price of the asset, buy the asset and take a short position in the forward contract. At the maturity of the contract the spot instrument is sold at the agreed price and the proceeds from the sale used to pay off the loan which has now grown to $S(t)e^{r(s-t)}$. The difference between the agreed amount and the borrowing is therefore a guaranteed profit.

If, on the other hand, the market forward price is less than the carry relationship ($F(t, s) < S(t)e^{r(s-t)}$), investors can sell the asset at the spot price, depositing the

¹ Leong (1997) pp 133.

proceeds in a bank account yielding r and simultaneously take a long position in a forward contract. At the maturity of the contract the asset is bought for F under the forward agreement using the proceeds from the initial deposit, once again yielding a profit equal to the difference between the market price and the cost of carry level.

The cost of carry relationship is only slightly more complicated if the asset pays a continuous yield δ . Recall that the resulting futures pricing formula is given by equation (1.1). The above arbitrage arguments hold here as well once the continuous dividend yield accruing to the physical position is taken into account.

4.2.2 Forward Price Bounds for Energies

The cost of carry relationship holds in markets where arbitrageurs are able to easily purchase and short sell assets. Now consider the cost of carry relationship for an energy, say oil. For the short future, long spot arbitrage, the arbitrageur would need to pay to store the commodity and transport the energy to the buyer of the forward at the maturity of the contract. The financing cost therefore now consists of the money market interest rate plus the cost of storage. This suggests a cost of carry relationship of the form

$$F(t, s) = S(t)e^{(r+w)(s-t)} \quad (4.3)$$

where w represents the cost of storage on a yield basis. Both the interest rate and the cost of the storage are positive and so the futures price is greater than the spot level implying an upward sloping (contango) forward curve, and this indeed corresponds to the observed curve for crude oil at the beginning of March 1999. However, often the forward curve is downward sloping (backwardation) as can be seen by the market curve at the end of March 2000 in Figure 4.1. This implies that the lower bound on forward pricing is not being met with the cost of carry described here not the only determinant of the futures price. One argument regularly used is that it is often not possible to sell the spot energy short (for example by leasing) and therefore, since the arbitrage cannot be executed, there is nothing to stop forward curves being in backwardation.

The current discussion implies that, in times of backwardation, market participants, instead of buying oil and storing it for future use, should purchase oil for future delivery at a discount to the spot price. However, for some participants holding the spot energy has value relative to having a forward contract. The benefit derived has been termed the *convenience yield*² or consumption value as companies keep the energy in inventory in order to be able to consume it if they want. In the oil industry inventory holders such as end users have a high convenience yield since they cannot afford to be short of physical oil.

We can represent the convenience yield therefore in terms of an effective continuous dividend stream which the holder of the spot will receive and which we again denote by δ .

² Brennan (1989) defined the convenience yield as the "...flow of services which accrues to the owner of a physical inventory but not to the owner of a contract for future delivery".

The full cost of carry for an energy therefore becomes;

$$F(t, s) = S(t)e^{(r+w-\delta)(s-t)} \quad (4.4)$$

Depending on the relative size of the storage costs and convenience yield the resulting forward curve can be in contango or backwardation. Gabillion (1995) discusses the limit to contango and backwardation in terms of the levels of storage and financing costs and on the convenience yield.

4.2.3 Seasonality in Prices

The forward curve for crude oil is downward sloping most of the time, but as we have seen can move in to contango, sometimes very quickly. Forward curves such as natural gas, however, cannot be so easily described, often exhibiting strong seasonal cycles. An example of this is the NYMEX natural gas curve illustrated in figure 4.1. For this curve, the peaks occur during the northern hemispheres winter when natural gas is in high demand for heating.

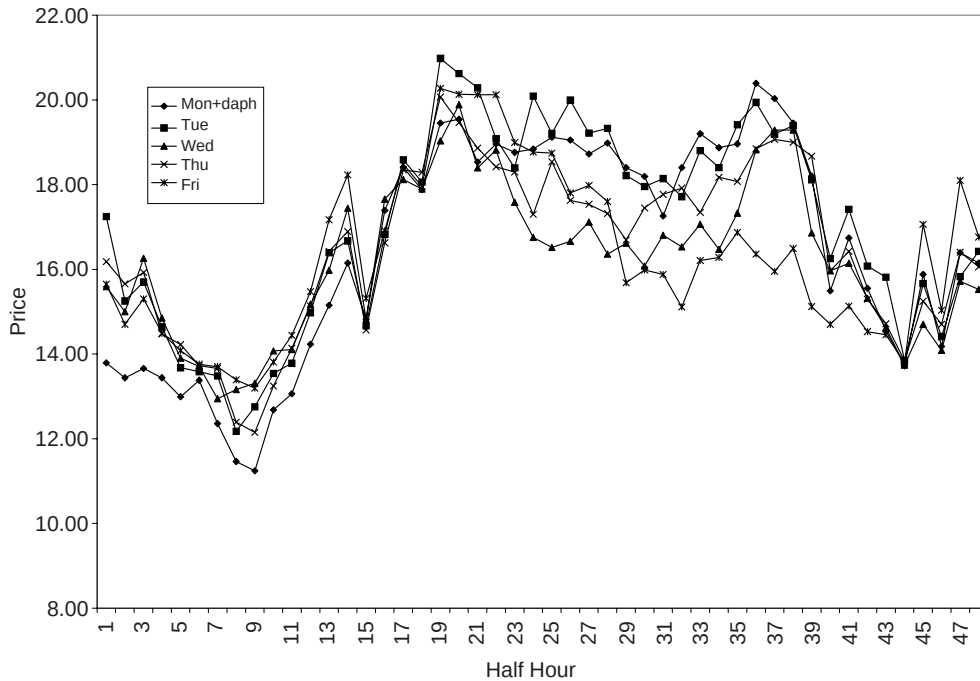
The size of the difference between the low point on the gas curve (\$2.69 for the April 2001 contract) and the high point the following winter (\$2.98 for the December 2001 contract) suggests that there is a clear arbitrage opportunity – buy natural gas forward when prices are low and store the gas until the next winter peak. Execution of this arbitrage should continue until the peaks disappear. Leong (1997) suggests that the presence of arbitrage opportunities of this kind continue to exist because of the inefficiency of the gas forward market compared to the financial markets. He also suggests several reasons for this inefficiency;

- there is little incentive of many players in the gas market to be cost conscious as increases in gas prices can be passed on to end users by many utilities.
- in order to be able to execute arbitrage a player needs a presence in both the spot and derivative markets – for example, the arbitrage described above requires the user to buy and store gas.
- this kind of relative value trading is new to many gas market players and, as the market becomes more efficient, this kind of discrepancy will diminish.

These arguments suggest that as the players in the market become more familiar with arbitrage between spot and forward markets and more conversant with derivatives the level of seasonality should diminish.

4.3 Forward Curves in the Electricity Market

Of all energy commodities electricity exhibits the most complicated seasonal patterns. In many markets there are two peaks in the forward curve as a result of weather. A peak is observed in winter due to the reliance on heating and again in summer because of air conditioning. Additionally, there can exist daily and hourly patterns. Figures 4.2(a) and 4.2(b) show spot price seasonality for the Australian electricity market for weekdays and



(Note: Monday and day after public holiday (daph) has been grouped together)

FIGURE 4.2(a) Median Price For Weekdays

weekends respectively. The figures show the median price per MWh for each of the 48 half hours for the different days for the NSW market.

Leong (1997) suggests a number of reasons for this strong level of seasonality. Firstly the more complicated patterns are due to the non-storability of electricity – once generated it has to be consumed immediately. This implies that arbitrage relationships which depend on storing the energy breakdown once the electricity has been generated although importantly electricity can be stored indirectly via the generator fuel³. The difficulty of storing electricity means that there is an absence of a leasing market which affects the short spot energy, long forward arbitrage.

A third suggested reason is that electricity is still a very regional market. Differences between regions arise due to differing methods used to produce electricity, weather patterns, demographics, local supply and demand conditions, etc. Links between regions, such as that between New South Wales and Victoria in Australia, help to reduce such regional variations, but constraints on the lines mean that they don't disappear completely.

As with other forward markets the forward curve should be constructed to fit available market prices. Unfortunately the illiquidity of many instruments in the

³ Electricity can also be stored directly via hydroelectric schemes.



(Note: Sunday and public holiday (ph) has been grouped together)

FIGURE 4.2(b) Median Price Weekends

electricity area means we often have to turn to other techniques. The current literature is dominated by three main approaches to building forward curves in the electricity market and we briefly describe the approaches in the following sections. The *arbitrage approach* relates the forward electricity curve to the prices of the fuels used to generate the electricity. The *econometric approach* uses econometric techniques to derive the forward prices of electricity by taking account of key econometric variables such as fuel cost, weather patterns, etc. Finally what we will call the *spot price modelling approach* derives forward prices by modelling underlying stochastic factors such as the spot price, convenience yield, long term mean price, etc. and deriving analytical formulas to characterise the curve.

4.3.1 Arbitrage Pricing Approach

Although electricity cannot be easily stored the fuels used to generate the electricity can be stored, and the link between the two fuels implies that the forward curve for electricity should be related to the input fuels. The arbitrage pricing approach takes this into account by considering the conversion process.

One of the key steps in the conversion process is the generation process itself and this depends on the efficiency of generation expressed as the *heat rate* – the number of BTU's (British Thermal Unit) required to generate one kWh of electricity. A basic electricity forward curve can be obtained from the fuel forward curve via the following relationship:

$$Cost_{electricity} = Heat\ Rate \times Price_{fuel} \quad (4.5)$$

A constant value of the heat rate implies that the shape of the electricity forward curve should resemble the forward curve of the input fuel. The calculation of the heat rate, especially for long term contracts is notoriously difficult given the speed of technological innovation in the electricity industry. The cost of electricity can be converted into a forward price after taking into account costs associated with fixed assets, transmission and tolling charges and others such as fuel purchase, fuel storage, and fuel transportation. These costs obviously change through time.

4.3.2 The Econometric Approach

Griswold (1997) discusses a simulation model which has inputs such as capacity ratings of transmission lines, fuel costs, total load by sub-region on an hourly basis, and data relevant to adjacent regions to compare incremental power prices. This type of analysis is performed by a number of commercial software packages for forward curve calculation. The output to a model of this type, however, is a prediction or forecast of spot hourly energy prices in the future. It is not a forward price in the sense of our definition in section 4.1 – i.e. the price which we can lock into today for trading the spot instrument at some time in the future.

4.3.3 The Spot Price Modelling Approach

This approach derives forward energy prices from assumptions about the stochastic factors driving the spot energy price and other key variables (for example, the long term price, the convenience yield, or interest rates). Models of this kind are well known in the fixed income market. For example, traditional models of interest rates, such as Vasicek (1977) and Cox, Ingersoll, and Ross (1985) explain the whole term structure of interest rates (discount bond prices, spot and forward rates) in terms of a stochastic description of the short term interest rate⁴. The approach involves specifying the stochastic factors that drive the evolution of the spot price. Forward prices, and all other derivatives whose value depends on the stochastic factors and time, can then be derived after applying various mathematical techniques. Depending on the assumptions employed about the random driving factors futures prices are often derived analytically. Examples of this

⁴ Multi-factor interest rate models such as Longstaff and Schwartz (1992) and Fong and Vasicek (1992) attempt to explain the term structure by modelling extra stochastic factors such as short rate volatility, the long rate, and other factors.

approach include Gabillion (1995) and Pilipovic (1997) who both suggest modeling the spot energy price and a long term price. All the models we study in chapter 6 can be viewed as being consistent with this approach.

4.4 Summary

In this chapter we have discussed the importance of forward curves for risk management and derivative pricing in the energy markets. Forward curve construction is more difficult in the energy markets when compared to the financial markets because of the cost of storing the spot energy, the fact that some energies derive a convenience yield to the holder, and the illiquidity of forward instruments in many markets.